Devs House 26

ZK-Solvency: Continuous Private Proof of Reserves

Taskforce

Blockchain & dApps and Fintech

GDG on Campus

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Problem Statement

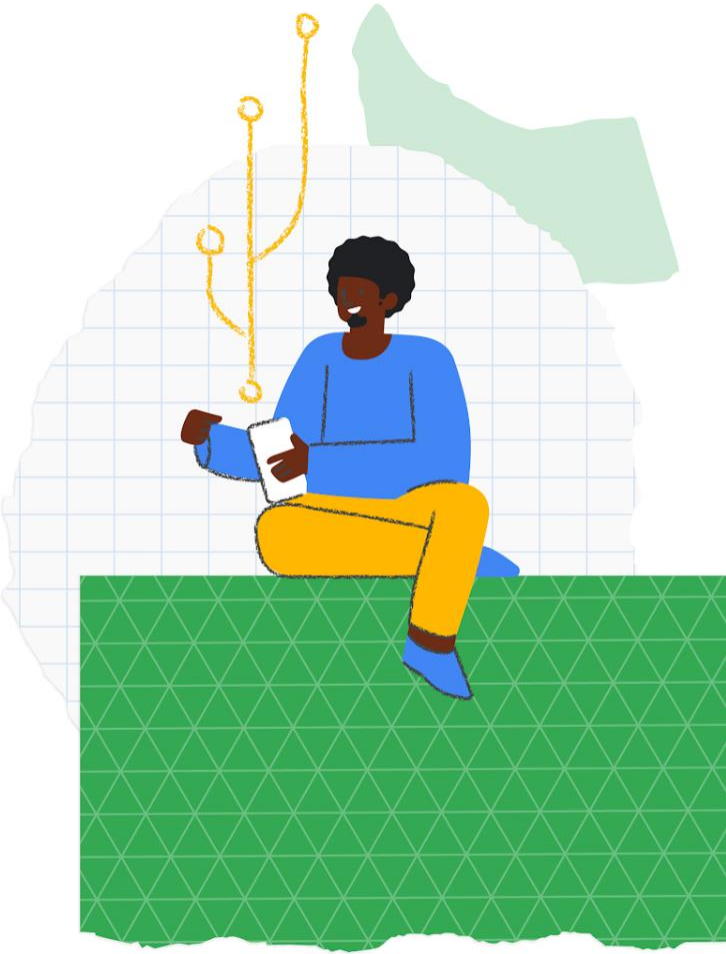
The Problem With Trusting a Crypto Exchange

- When you deposit funds on a centralized exchange, you're taking their word for it. But how do you actually know the money is there? Proving assets is easy — blockchain wallet signatures handle that. Proving liabilities is the hard part. It means verifying user balances, which raises an immediate problem: publishing that data violates privacy laws, exposes sensitive customer information, and reveals proprietary trading metrics.
- Most proof-of-reserve systems avoid this entirely. They run monthly snapshots, verify assets, and stop there — leaving room for hidden negative balances, omitted accounts, and window-dressing fraud to slip through undetected. A solvent exchange at audit time can be insolvent an hour later. And nothing stops an exchange from quietly excluding thousands of small accounts to artificially shrink their reported liabilities.
- Even when exchanges want to do this properly, they control the data going into the proof — which means a cryptographically valid proof over dishonest inputs is still a lie dressed in mathematics. Users are left trusting systems that were never built to be fully transparent, audited only when the exchange chose to be, and verified only by people the exchange invited to look.

Product Idea

How We Solve It

- We built a Zero-Knowledge Proof of Solvency system using zk-STARKs — proving an exchange is solvent without revealing a single user balance or proprietary number.
- Before any computation begins, zkTLS attestation ensures the exchange cannot manipulate inputs — liability data is cryptographically verified straight from their authenticated internal system, untouched. Every balance is then hashed into a Merkle Tree and committed against an on-chain Nullifier Registry, making silent account omission mathematically impossible. The zk-STARK proof guarantees three things simultaneously: every account is included, no negative balances exist, and total assets exceed total liabilities.
- Proofs run every ten minutes on a decentralized prover network and are cryptographically chained — a gap is on-chain evidence of manipulation. Every user gets a one-click inclusion proof confirming their balance was counted. Regulators get a signed circuit registry with credentialed audit attestations for full compliance.
- Cryptographic honesty, at scale, continuously, without compromising privacy



Additional Info

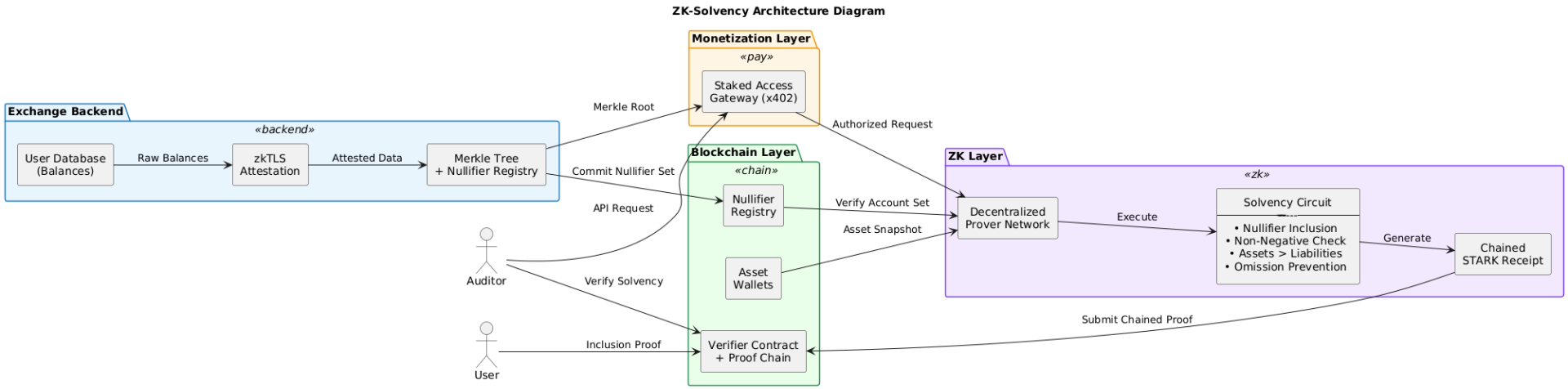
zkTLS Attestation + Nullifier Registry

Decentralized zk-STARK Prover Network

Exchange Backend + On-Chain Wallets + Smart Contract Verifier

Real-Time, Privacy-Preserving, Cryptographically Verifiable Solvency

Architecture Diagram



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How Unique is it?

Existing Work vs Coined Novelty

- Periodic snapshot audits (not real-time)
- No cryptographic proof of total liabilities
- Vulnerable to hidden negative balances
- Silent account omission undetectable
- Mostly asset-side verification only

- zk-STARK proof of solvency
- (Assets > Liabilities)
- zkTLS-attested inputs, tamper-proof by design
- Enforced non-negative balance constraints
- On-chain Nullifier Registry prevents omissions
- Continuous chained proofs, not snapshots

Tech Stack

- Cairo (zk-STARK) + Solidity
- PostgreSQL + Merkle Tree + Nullifier Registry
- Staked Access Gateway (x402) + zkTLS Attestation
- Decentralized Prover Network (Gevulot / Succinct SP1)
- Basel Proofs • Stablecoin Audits • RWA Verification

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